NOMAN AHMAD

Backend Engineer | Python, FastAPI & AI Systems
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Summary

B.Tech Computer Science student graduating in 2027 with backend internship experience building 40+ FastAPI/PostgreSQL endpoints, multi-role RBAC, and integrated business modules. Experienced in database migrations, real-time systems, and deployed applications. Builds reliable LLM workflows using structured outputs and deterministic decision logic.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, SSE, JWT, RBAC, Pydantic
Frontend & Mobile: React, Next.js, Tailwind CSS, Flutter
Databases: PostgreSQL, MongoDB, SQLAlchemy, Alembic, pgvector
Testing & DevOps: Pytest, Ruff, mypy, Git, GitHub Actions, CI/CD, Docker, Render
AI / ML: Google Gemini, LLM orchestration, structured outputs, RAG, embeddings, FAISS, YOLO, Roboflow

Experience

Software Developer Intern, Dream Filler *Sep 2026 – Present*

- Fine-tuned YOLO across 4 iterations on a self-collected, self-labeled 62-image dataset using Roboflow; achieved 75.5% mAP and 11.8 ms inference latency.
- Introduced a dedicated Patient model, migrated existing MongoDB data, and adapted appointment workflows to use patient-linked records.
- Built custom Flutter camera capture with image compression, replacing `image_picker` to address memory-related app restarts.
- Implemented Server-Sent Events (SSE) for server-pushed, real-time queue dashboard updates, reducing reliance on repeated polling.
- Built a public React appointment interface and added or modified 15+ Node.js/Express API endpoints across 3–5 modules.

Backend Development Intern, Quantumbay Cloud (Remote) *Jul 2026 – Sep 2026*

- Built 3 modules and 40+ FastAPI/PostgreSQL endpoints for Medorax, a pharmacy ERP; integrated with 10+ modules and the frontend.
- Implemented RBAC for owner, manager, accountant, pharmacist, and inventory manager roles with per-module and branch-level permissions.
- Designed schemas and Alembic migrations; resolved migration conflicts across independently developed modules.

Projects

AI Requirement Analyzer & Decision Engine *FastAPI, Next.js, PostgreSQL, Gemini* [Code]

- Turns vague client specs into structured requirements: Gemini extracts, a deterministic engine decides TAKE / REVIEW / DECLINE.
- Enforced schema-validated JSON (Pydantic v2) with automated recovery from malformed LLM output.
- Bounded context assembly with provenance tags to reduce hallucination; multi-tenant workspaces and capability matching.

AlumniConn — Student–Alumni Platform *React, TypeScript, FastAPI, PostgreSQL, pgvector* [Code] [Live]

- Built a deployed multi-tenant platform with tenant-scoped data access, four-role RBAC, and JWT authentication.
- Implemented hybrid mentor recommendations using 384-dimensional embeddings, pgvector similarity search, and deterministic ranking heuristics.
- Developed real-time WebSocket messaging with online presence, persistent chat history, and Cloudinary image attachments.
- Added a one-click demo environment with synthetic institutional data and an in-browser email verification inbox.

RepoMind — Repository Intelligence CLI *Python, Tree-sitter, pgvector, Gemini* [Code]

- Combined deterministic Tree-sitter analysis with Gemini reasoning to surface technical debt; CI with Pytest, Ruff, mypy, GitHub Actions.

AI Ticket Triage *Python, Gemini, SQL, REST* [Code]

- Pipeline normalizes tickets, detects language, extracts IDs, checks VIP status, and flags legal-risk terms before any LLM call.
- Gemini handles only intent, sentiment, and routing, with a human-in-the-loop dashboard for editable AI drafts.

Achievements & Certifications

India Runs Data & AI Challenge — FAISS + Sentence-Transformers ranking pipeline for 100,000+ resumes, deployed on Hugging Face Spaces. [Code] [Demo]

Certifications: DSA using Java, NPTEL (Nov 2025); Responsive Web Design, freeCodeCamp (Aug 2025)

Education

Vindhya Institute of Technology and Science Satna, Madhya Pradesh
Bachelor of Technology (Computer Science) **Expected 2027**